NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

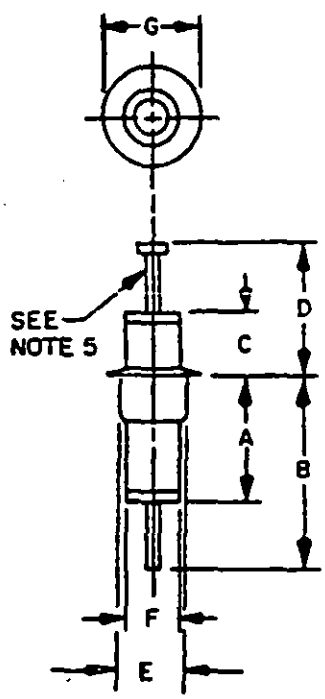
MIL-PRF-15733/62C
26 November 1982
SUPERSEDING
MIL-F-15733/62B
25 July 1979

PERFORMANCE SPECIFICATION SHEET

FILTERS, RADIO FREQUENCY INTERFERENCE, STYLES FL31 AND FL47

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the filters described herein shall consist of this specification and the latest issue of MIL-F-15733.

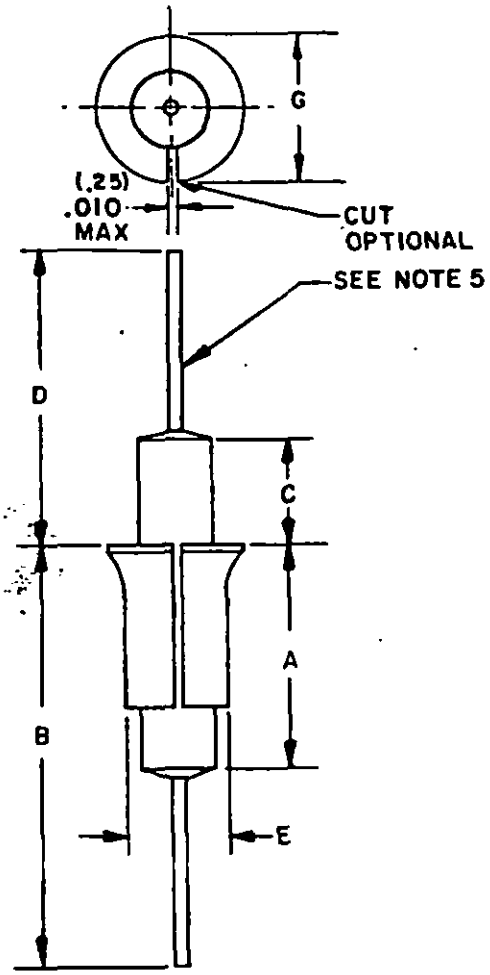


Dash No. 0001

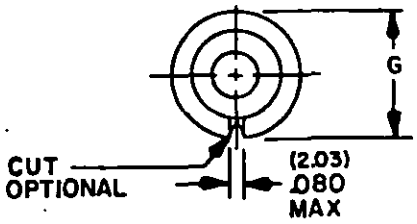
FIGURE 1. Case dimensions and circuit configuration.

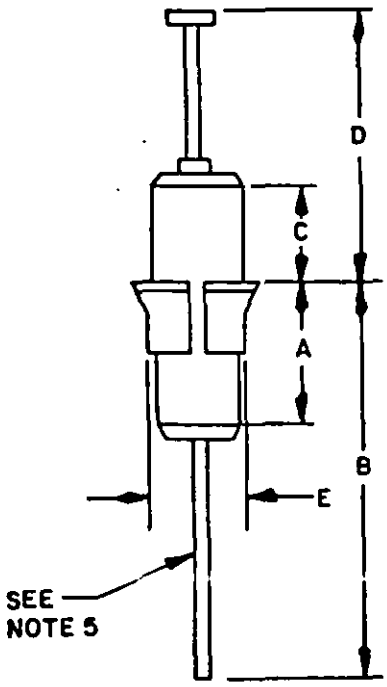
FSC 5915

Source: https://assist.dla.mil -- Downloaded: 202610-18T07:28Z
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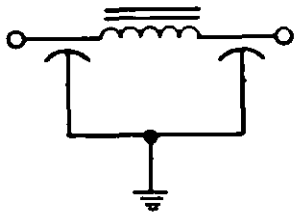


Dash Nos. 0002 and 0005





Dash Nos. 0003 and 0004



TYPICAL CIRCUIT DIAGRAM

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
4. Circuit diagram is for information only.
5. Leads shall be solid, tinned, AWG 20, .032 (.81 mm) diameter.

FIGURE 1. Case dimensions and circuit configuration - Continued.

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TABLE 1. Dimensions.

Type designation	Dash no.	A		B		C		D		E		F		G	
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
FL31	0001	.219 (5.56)	.281 (7.14)	.375 (9.52)	.437 (11.10)	.125 (3.18)	.187 (4.75)	.281 (7.14)	.343 (8.71)	.130 (3.30)	.160 (4.06)	.110 (2.79)	.140 (3.56)	.175 (4.44)	.205 (5.21)
FL47	0002	.185 (4.70)	.277 (7.04)	1.154 (29.31)	1.308 (33.22)	.078 (1.98)	.140 (3.56)	1.047 (26.59)	1.171 (29.74)	.095 (2.41)	.125 (3.18)	---	---	.133 (3.37)	.153 (3.89)
FL47	0003	.247 (6.27)	.297 (7.54)	.375 (9.52)	.500 (12.70)	.141 (3.58)	.171 (4.34)	.281 (7.14)	.343 (8.71)	.130 (3.30)	.160 (4.06)	---	---	.175 (4.44)	.205 (5.21)
FL47	0004	.131 (3.33)	.171 (4.34)	.749 (19.02)	.811 (20.60)	.078 (1.98)	.108 (2.77)	.266 (6.76)	.281 (7.14)	.130 (3.30)	.160 (4.06)	---	---	.188 (4.77)	.203 (5.16)
FL31	0005	.284 (7.21)	.346 (8.79)	1.284 (32.61)	1.365 (34.67)	.094 (2.39)	.125 (3.18)	1.094 (27.59)	1.156 (29.36)	.123 (3.12)	.153 (3.87)	---	---	.160 (4.06)	.190 (4.82)

FIGURE 1. Case dimensions and circuit configuration - Continued.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Weight:

Dash number 0001 - 1.2 grams.

Dash numbers 0002, 0003, and 0004 - .7 gram.

Dash number 0005 - .9 gram.

Case: Not applicable (metallic mounting eyelet termination).

Terminals, mounting termination: Solderable.

Operating temperature range: -55°C to +125°C.

Rated voltage: See table II, voltages given are over the rated temperature range of -55°C to +125°C.

Rated current: 10 amperes, dc or ac (rms).

Voltage conditioning (quality conformance inspection only): The test shall be performed as follows:

Test temperature: 125°C +4°C, -0°C.

Test voltage: 200 percent of the rated dc voltage. Charging current shall not exceed 50 mA.

Points of applications of test voltage: Between each terminal and case.

Duration of exposure to test voltage: 100 ±4 hours.

After completion of exposure, and while filters are established at 125°C, insulation resistance shall be measured. Filters shall then be stabilized at 25°C, and insulation resistance shall be measured. A reject shall be defined as a filter whose insulation resistance does not meet initial requirements when measured at 125°C and 25°C conditions as specified.

Seal: Not applicable.

Solderability: MIL-STD-202, method 208.

Number of terminals to be tested - 2.

Capacitance to ground: MIL-STD-202, method 305. Minimum capacitance shall be as specified in table II.

Test frequency: 1,000 ±100 Hz.

Temperature rise: 25°C, maximum.

Dielectric withstanding voltage: MIL-STD-202, method 301.

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TABLE II. Electrical characteristics.

Type designation	Dash number	Rated voltage		Capacitance (min)	Minimum insertion loss (dB) in accordance with MIL-STD-220																							
					+25°C 1/												-55°C and +125°C 1/											
		V dc	2/ Vrms	pF	5 MHz	10 MHz	20 MHz	50 MHz	100 MHz	200 MHz	400 MHz	500 MHz	1 GHz	2 GHz	10 GHz	5 MHz	10 MHz	20 MHz	50 MHz	100 MHz	200 MHz	400 MHz	500 MHz	1 GHz	2 GHz	10 GHz		
FL31	0001	200	140	1,500	--	3	13	32	45	58	--	70	70	--	70	--	--	5	15	28	46	--	70	70	--	70		
FL47	0002	100	--	1,500	--	5	12	28	45	--	--	68	70	70	--	--		5	15	28	--	--	68	70	70	--		
FL47	0003	70	--	1,500	5	--	19	--	50	--	65	--	65	--	65	--	--	7	--	32	--	63	--	65	--	65		
FL47	0004	70	--	12,000	--	25	40	--	70	--	65	--	--	--	65	--	11	17	--	70	--	63	--	--	--	65		
FL31	0005	200	140	3,000	3	12	23	38	50	60	--	--	65	65	65	--	5	10	16	30	48	--	--	65	65	65		

1/ Full-load insertion loss measurements shall be performed at 5, 10 and 20 MHz, all other measurements shall be performed at no-load.

2/ 0 to 400 Hz.

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The test voltage shall be applied for 1 to 5 seconds (by dash numbers):

0001 and 0005 - 1,000 V dc

0002 - 600 V dc

0003 - 300 V dc

0004 - 200 V dc

Charging current shall not exceed 50 mA.

Barometric pressure (reduced) (by dash numbers): MIL-STD-202, method 105.

0001 - test condition D (100,000 ft.)

0002 and 0005 - test condition E (150,000 ft.)

0003 and 0004 - test condition B (50,000 ft.)

Dielectric withstanding voltage: In accordance with initial requirements, except that the test voltage shall be 1.2 times the rated dc voltage.

Insulation resistance: MIL-STD-202, method 302, test condition A. Minimum insulation resistance measured between both terminals connected together and the case (by dash numbers):

	At 25°C	At 125°C
0001, 0002, and 0005	10 gigohms	1 gigohm
0003 and 0004	5 gigohms	500 megohms

Voltage drop: 0.1 volt.

Insertion loss: In accordance with MIL-F-15733 and table II.

Overload: In accordance with MIL-F-15733.

Measurements at 25°C after test: Insulation resistance shall meet initial requirements.

Transient voltage (dash numbers 0002 and 0003 only): Filters shall withstand the transient and spike voltages specified in MIL-STD-704.

Measurements at 25°C after test: Insulation resistance shall be measured and shall meet initial requirements.

Terminal strength: MIL-STD-202, method 211, test condition A.

Applied force: 5 pounds.

Flashpoint of impregnant or potting compound: Not applicable.

Resistance to soldering heat: MIL-STD-202, method 210, test condition B.

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Resistance to solvents: Not applicable.

Salt spray (corrosion): Not applicable.

Thermal shock and immersion: Not applicable.

Shock (specified pulse): MIL-STD-202, method 213, test condition I.

Electrical load: 100 percent of the rated voltage between terminals and case.

Measurements after test: Insertion loss (check test) shall meet initial requirements.

Vibration, high frequency: MIL-STD-202, method 204, test condition D.

Electrical load: Rated voltage and rated current shall be applied continuously during test.

Test during vibration: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent-open or short-circuiting.

Moisture resistance: MIL-STD-202, method 107, test condition A, except that the temperature of step 3 shall be 125°C +3°C, -0°C. No measurement shall be made before and after cycling.

Filters shall then be tested in accordance with MIL-STD-202, method 106.

Initial measurements and load voltage: Not applicable.

Polarization voltage: Not applicable.

Measurements after 24 hours drying period, at a temperature not to exceed 85°C, and a relative humidity of 50 percent:

Dielectric withstanding voltage with 90 percent of initial voltage applied for 5 ±1 seconds.

Insulation resistance shall be not less than 1 gigohm.

Insertion loss (check test) shall meet the initial requirements.

Life: MIL-STD-202, method 108, test condition D.

Test temperature: 125°C.

Test voltage (by dash number):

0001 and 0005 - 1.2 times the rated rms voltage.

0002, 0003, 0004 - 1.4 times the rated dc voltage.

Test current: 10 amperes.

Measurements after test:

Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds.

Insulation resistance shall be not less than 1 gigohm.

Insertion loss (check test) shall meet the initial requirements.

Marking: Filters shall not be marked. Full marking, in accordance with MIL-F-15733 shall be marked on the unit package.

Part number: M15733/62 (dash number from table I).

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Supersession data: Filters in this specification supersedes filters in MIL-F-15733/37(USAF) and MIL-F-15733/50(USAF) as follows:

Part number MIL-F-15733/62	Superseded part number
0001	M15733/37-0001
0002	M15733/50-0001
0003	M15733/50-0002
0004	M15733/50-0004

QUALITY ASSURANCE PROVISIONS:

Qualification insection: In accordance with MIL-F-15733, except that the transient voltage test shall be performed as an additional test in group I, following the overload test, for dash numbers 0002 and 0003 only.

Quality conformance inspection: In accordance with MIL-F-15733, except as follows:

Group A: In accordance with MIL-F-15733, except that prior to the group A inspection 100 percent of each lot of filters shall be subjected to the voltage conditioning test. If the total rejects from any particular lot exceed 10 percent that entire lot shall be rejected.

Group B: In accordance with MIL-F-15733, except that the transient voltage shall be performed as an additional test, following the overload test, for dash numbers 0002 and 0003.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, GL
Navy - SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME, AT, AV, WC
Navy - CG, MC, AS, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5915-U244-2)

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